Tariq Khalaf

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Education

University of North Carolina at Charlotte **Expected: Dec 2028**
Bachelors of Science in Electrical and Computer Engineering

Skills

Design & Simulation: KiCad, Fusion 360, AutoCAD, SolidWorks, PCB Design, Circuit Diagrams, VS Code, Dev-C++, Meshmixer, 3D Printing.
Programming: Python, C++, JavaScript, Arduino, Embedded C, UART.
Engineering Skills: Soldering, PCB bring-up/debugging, USB-C power/programming, brushed DC and servo motor control, custom boards, circuit prototyping.

Experience

Undergraduate Research Assistant **Aug 2026 – Present**
University of North Carolina at Charlotte

- Conducting research on SLAM (Simultaneous Localization and Mapping) for indoor positioning applications.
- Exploring RSSI-based signal analysis and trilateration techniques to estimate device location; project is ongoing.

Teacher's Assistant **Jan 2026 – Jul 2026**
Central Piedmont Community College

- Assisted students during labs across 15+ projects, teaching hands-on techniques, equipment navigation, and troubleshooting builds.
- Supported hands-on work in electronics, coding, and mechanical/civil engineering builds using Arduino, circuits, sensors, and prototypes.

Projects

Brushed DC Electronic Speed Controller – Academic Project **Dec 2025 – Jan 2026**

- Designed a 24V DC motor ESC supporting up to 5A continuous current, emphasizing safe power delivery and reliability.
- Created multi-layer PCB layouts in KiCad with ground planes, power pours, via stitching, and high-current trace planning.
- Selected parts using datasheets and supplier availability to balance performance, cost, footprint, and manufacturability.

Embedded MCU Power & Programming Board – Personal Project **Jun 2026 – Jul 2026**

- Designed a custom MCU board accepting 24–36V input, stepped down to 3.3V using an onboard buck regulator.
- Integrated USB-C for board power/programming with required power, programming, and protection support circuitry.
- Planned PCB layout around regulator placement, decoupling, grounding, and clean 3.3V delivery for embedded hardware.

Automated Drawbridge on a Custom MCU Board – Academic Project **Apr 2026 – May 2026**

- Built a motorized bridge that opens and closes automatically, running off a custom MCU board driving one brushed DC motor and a servo through a TB6612FNG dual H-bridge driver.
- Automated crossing logic from ultrasonic sensor input, with traffic LEDs and buzzers signaling bridge state to vehicles and boats.
- Fabricated the structure from a self-designed 3D-printed bridge deck on hand-cast cement pillars, integrating mechanics, power distribution, and control into one working prototype.

Remote-Control Robot – Personal Project **Jan 2026 – Feb 2026**

- Built an ESP32 robot driving two brushed DC motors through a TB6612FNG dual H-bridge with wireless remote input.
- Integrated an ultrasonic sensor, motion detector, buzzers, and an onboard camera for obstacle detection and alerts.
- Wired power distribution and common grounds across driver, MCU, and sensors; tuned PWM drive logic for stability under load.

Custom Claude AI Interface – Personal Project **Jul 2026 – Aug 2026**

- Built a custom graphical user interface (GUI) for Anthropic's Claude AI, giving users a simpler, more streamlined way to interact with the assistant.
- Wrote a set of Markdown preference files that let each user tailor how Claude responds to them, including tone, response length, formatting, and recurring workflows, without changing any code.

Synchronous Buck Converter – Robotics Club Project **Oct 2026 – Present**

- Designing a synchronous buck converter that regulates a 4S LiPo battery input to a stable 12V output rail, built to spec at the request of the robotics club's electrical lead.
- Chose a single-phase topology for a power stage designed to handle up to 50A of load current.
- Laying out a 4-layer, 1 oz copper PCB with via stitching between layers to carry high-current paths and keep a solid ground.

Involvements

Charlotte Rocketry Club – UNC Charlotte **2026 – Present**

- New member contributing to PCB design for club projects as the semester's build cycle gets underway.

49er Robotics – UNC Charlotte **2026 – Present**

- Designing PCBs for club projects, including a synchronous buck converter power board.